

ROHIT NEGI

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PROFESSIONAL SUMMARY

B.Tech (Computer Science & Engineering) student at Haridwar University with hands-on industry internship experience in Machine Learning, Explainable AI (XAI), and full-stack model deployment. Proficient in Python, Scikit-learn, TensorFlow/Keras, XGBoost, SHAP, Flask, Docker, and Tableau, with demonstrated experience building end-to-end ML platforms, recommendation systems, and interactive BI dashboards. Seeking a Machine Learning Engineer or software engineering role.

EDUCATION

Haridwar University — B.Tech, Computer Science and Engineering	2023 - 2027 (Expected)
CGPA: 7.00/10.00 Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Computer Vision	
Senior Secondary (CBSE), St. John Public School Percentage: 83.0%	2021 - 2023
Secondary (CBSE), St. John Public School Percentage: 91.0%	2019 - 2021

WORK EXPERIENCE

AI & ML Intern — Aarsh AI Technologies	8 June 2026 - 25 July 2026
<i>New Delhi, India 2-Month Industry Internship</i>	

- Completed a 2-month industry internship in Artificial Intelligence & Machine Learning, gaining intensive hands-on experience in production AI/ML development and problem-solving.
- Developed an Intelligent Employee Attrition Prediction & Retention Platform to assess employee resignation risk and empower data-driven HR retention decisions.
- Built an end-to-end machine learning pipeline using Python and Scikit-learn, covering data preprocessing, feature engineering, model training, prediction, and evaluation.
- Implemented SHAP-based Explainable AI (XAI) to interpret model predictions and identify key organizational and demographic factors contributing to employee attrition risk.
- Developed a Flask-based web application to integrate trained ML models into an interactive, user-facing dashboard for real-time risk scoring and recommendations.
- Deployed the platform live using Vercel, demonstrating an end-to-end production workflow: ML Model Development → Explainability (XAI) → Web Integration → Serverless Deployment.

Machine Learning Intern — Prodigy InfoTech	1 Feb 2025 - 28 Feb 2025
<i>Remote, India 1-Month Virtual Internship</i>	

- Successfully completed a 1-month intensive Machine Learning internship from 1st February, 2025 to 28th February, 2025, receiving outstanding remarks.
- Worked across the end-to-end machine learning pipeline, including data preprocessing, feature engineering, exploratory data analysis, model training, and performance evaluation on structured datasets.
- Implemented, fine-tuned, and evaluated supervised learning algorithms in Python utilizing Scikit-learn and Pandas, benchmarking performance metrics (Accuracy, Precision, Recall, F1-Score) against baseline models.
- Completed comprehensive Machine Learning with Python coursework and practical projects sponsored by Prodigy InfoTech with hands-on assessments.

PROJECTS

Intelligent Employee Attrition Prediction & Retention Platform | Python, Flask, Scikit-learn, SHAP, Vercel

- Architected and deployed a full-stack ML platform on Vercel predicting employee turnover and delivering explainable, personalized retention strategies.
- Trained and benchmarked 8+ ML algorithms (XGBoost, LightGBM, ExtraTrees, Ensembles) with SMOTE and 5-fold CV, achieving 85.0% accuracy and 77.3% ROC-AUC.
- Optimized model deployability by 23x, distilling a 61MB ensemble into a 2.6MB calibrated model to cut cold-start deployment time from 15 min to ~1 min.
- Integrated SHAP for Explainable AI (XAI) feature attribution, demographic fairness checks, and an automated recommendation engine for HR retention planning.

IBM HR Analytics Interactive Dashboard | Tableau, Exploratory Data Analysis, Business Intelligence

- Designed an interactive Tableau dashboard tracking 1,470 employee records to evaluate workforce demographics, retention rates, tenure (avg. 7.0 yrs), and experience trends (avg. 11.3 yrs).
- Developed multidimensional visual analytics including treemaps, heatmaps, and dual-line charts to analyze gender distribution (60% Male / 40% Female), marital status, and age distributions across roles.

- Analyzed monthly income metrics across 9 job roles (\$2.6k to \$18k+) and education disciplines (600+ Life Sciences, 460+ Medical), surfacing compensation disparities to inform HR retention strategies.

Spotify Music Recommendation System | Python, Streamlit, Scikit-learn, Spotipy, Docker

- Developed a content-based music recommender using K-Nearest Neighbors (Scikit-learn) on Spotify audio features (danceability, energy, tempo, valence) to find the most similar tracks.
- Integrated the Spotify Web API (Spotipy) to enrich recommendations with live album artwork, 30-second audio previews, and direct Spotify web links.
- Built an interactive Streamlit web application with fuzzy song search (difflib), adjustable recommendation counts (3–10 tracks), and real-time inference.
- Containerized the application with Docker & Docker Compose, serializing model pipelines via pickle for reproducible container deployment on port 8501.

Cats vs Dogs Image Classification | Python, TensorFlow, Keras, CNN, OpenCV, NumPy

- Designed and trained a Convolutional Neural Network (CNN) in TensorFlow/Keras for binary image classification with data augmentation and dropout regularization.
- Preprocessed image datasets with resizing and normalization; tracked training and validation loss curves to optimize convergence and minimize overfitting.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML, CSS

Machine Learning & Deep Learning: Scikit-learn, XGBoost, LightGBM, SHAP (XAI), TensorFlow, Keras, CNN, OpenCV, Ensemble Methods

Data Analysis & Visualization: Pandas, NumPy, Plotly, Matplotlib, Seaborn, Tableau, Microsoft Excel

Tools & Platforms: Git, GitHub, Docker, Vercel, Flask, Streamlit, VS Code, Jupyter Notebook, Oracle Cloud (OCI), Linux

Core Competencies: Data Structures & Algorithms, Object-Oriented Programming (OOP), Model Distillation, System Design

CERTIFICATIONS

- Machine Learning Internship Certificate of Completion — Prodigy InfoTech
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate — Oracle
- Building RAG Agents with LLMs — NVIDIA
- CyberOps Associate — Cisco Networking Academy
- Machine Learning with Python — Prodigy InfoTech
- HTML, CSS, and JavaScript for Web Developers — Coursera

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **Data Structures & Algorithms:** Solved 100+ Coding Problems on LeetCode demonstrating strong analytical problem-solving and algorithmic skills.
- **3rd Position — Internal Smart India Hackathon (SIH) 2024:** Conducted by Haridwar University & Ministry of HRD / Institution's Innovation Council (IIC) on 14-09-2024; awarded Certificate of Recognition for developing innovative technological solution among competing university teams.
- **3rd Position — TechSangram (University-Level Competition):** Developed an IoT Smart Waste Management Bin prototype integrating automated sensor-based sorting and monitoring.
- **NPTTEL Certification with 70% Score:** Successfully achieved Elite certification; awarded academic merit scholarship of ₹1,100 twice by the institution.